🫁 Cardiorespiratory OSCE Preparation

Cases A & B — Community & Acute Cardiorespiratory Physiotherapy

Target: 70%+ (Very Good) | Physiotherapy OSCE

Case A — Mrs Fletcher (COPD) Case B — Ms Rodriguez (Asthma)

Mrs. Margaret Fletcher

CASE A

74F | Long-standing COPD | ↑ SOB & ↓ exercise tolerance (6 months) | 40 pack-year smoking

Barrel chest | Accessory muscle use at rest | Chronic productive cough (worse mornings) | Lives alone, difficulty with household chores

Reason for referral: Assessing suitability for community pulmonary rehabilitation

1 Subjective Assessment

"Hello Mrs. Fletcher, my name is [your name] and I'm a physiotherapist. Before we begin, I'd like to explain what today's assessment involves — I'll be asking you some questions about your breathing and daily activities, and later I may carry out some physical tests. Everything is confidential. Do you have any questions, and are you happy to proceed?"

KEY QUESTIONS TO ASK (6)

- **1. Can you describe your breathlessness?** At rest, on exertion, or both? How far can you walk on the flat before stopping? *— I ask this to establish her MRC Dyspnoea Grade, which directly determines pulmonary rehab eligibility and allows me to set a functional baseline.*
- **2. Tell me about your cough and sputum.** How much, what colour, has it changed recently? *— I ask this because chronic productive cough suggests sputum retention (ACBT may be indicated), and a colour change to yellow/green signals acute infection, which would contraindicate commencing rehab.*
- **3. How has your breathing affected your daily life?** Household chores, shopping, washing, dressing — what can you no longer do? *— I ask this to identify functional activity limitations, establish rehab goals, and assess risk given she lives alone.*

- **4. Do you still smoke, or have you stopped?** How many chest infections or hospital admissions have you had this year? *— Smoking cessation is the most effective disease modifier in COPD. Exacerbation frequency informs GOLD classification and prognosis.*
- **5. What medications are you currently taking, and can you show me how you use your inhaler?** *— Inhaler technique is poor in up to 80% of patients. I need to know she is optimally medically managed before adding exercise load.*
- **6. Do you have ankle swelling, morning headaches, or wake at night feeling breathless?** *— Morning headaches suggest nocturnal CO_2 retention (hypercapnia); ankle oedema + COPD raises the possibility of cor pulmonale — both are safety concerns that require medical review before rehab.*

RED FLAGS (5)

⚠ Haemoptysis

Blood in sputum in a 74-year-old with 40 pack-year history — must exclude malignancy, PE, or severe infection

→ Urgent GP/ED referral; do not commence rehab

⚠ SpO_2 <88% at rest

Severe resting hypoxaemia; exercise will worsen desaturation; may require supplemental O_2 assessment (LTOT)

→ Refer for ambulatory O_2 assessment; modify exercise intensity accordingly

⚠ Ankle oedema + central cyanosis

Suggests cor pulmonale (right heart failure from chronic hypoxaemia) — medical instability

→ Refer to physician before commencing any exercise programme

⚠ Sudden ↑ SOB or chest pain

Could indicate pneumothorax (high risk with emphysema), ACS, or PE — all life-threatening

→ Call emergency services immediately; do not proceed with assessment

⚠ Morning headaches / confusion

CO_2 retention causing hypercapnic respiratory failure; CNS effects of hypercapnia

→ Urgent medical review; ABG required; may need NIV

PROBLEM LIST

PHYSICAL IMPAIRMENTS	FUNCTIONAL ACTIVITY LIMITATIONS
Hyperinflation / barrel chest — ↓ diaphragm mechanical efficiency	Cannot complete household chores without stopping
Accessory muscle overuse at rest — ↑ work of breathing	Reduced walking distance (likely MRC Grade 3–4)
Chronic sputum retention — impaired mucociliary clearance	Difficulty with personal care (showering, dressing)

Peripheral muscle deconditioning — ↓ VO_2 max	Social isolation — limited community participation
Possible anxiety / dyspnoea fear-avoidance cycle	Falls risk and acute decompensation risk living alone

2 Pathophysiology

WHAT IS HAPPENING IN THE LUNGS

- COPD involves progressive, largely irreversible airflow limitation caused by a combination of **small airway disease** (obstructive bronchiolitis — narrowing and fibrosis of the small airways) and **parenchymal destruction** (emphysema — alveolar wall breakdown from chronic inflammatory proteolytic damage). Mrs. Fletcher's 40 pack-year history is the dominant aetiological driver.
- Emphysematous destruction causes **loss of elastic recoil**. Without the elastic tethering that normally holds airways open, **dynamic airway collapse** occurs during expiration — gas is trapped distally. Repeated air trapping leads to **chronic hyperinflation** — the barrel chest.
- The chronically hyperinflated, flattened diaphragm sits at a mechanically disadvantaged position (shorter muscle fibre length, suboptimal force-length relationship on its length-tension curve). Breathing therefore recruits **accessory muscles** — visible at rest in Mrs. Fletcher — massively increasing the energy cost of breathing.
- Smoke-damaged cilia and mucous gland hypertrophy cause **chronic mucus hypersecretion** with impaired mucociliary clearance. Sputum pools in dependent airways overnight (gravity-dependent), explaining why her cough is worst in the morning.
- Heterogeneous lung destruction creates severe **V/Q mismatch** — some alveoli ventilate but do not perfuse; others perfuse but do not ventilate. The net result is chronic hypoxaemia. Sustained hypoxaemia drives pulmonary arteriolar vasoconstriction → pulmonary hypertension → right ventricular strain (cor pulmonale).
- Systemic inflammation (elevated IL-6, TNF-α) combined with disuse from activity avoidance causes significant **peripheral muscle wasting and deconditioning** — contributing to exercise intolerance independent of lung function.

WHY MRS. FLETCHER FITS THIS PATHOPHYSIOLOGY

Barrel chest = chronic air trapping and hyperinflation from emphysema. Accessory muscle use at rest = diaphragm working at a mechanical disadvantage. Morning productive cough = overnight gravity-dependent sputum pooling with impaired clearance. Progressive 6-month decline in exercise tolerance = deconditioning layered onto worsening V/Q mismatch. Lives alone → fear-avoidance cycle limiting activity further.

3 Differential Diagnoses

DIFFERENTIAL 1: BRONCHIECTASIS

SUPPORTING FEATURES

- Chronic productive cough with large volumes of purulent sputum
- Recurrent chest infections / repeated antibiotic courses
- Coarse crackles in affected lobes on auscultation
- History of childhood whooping cough, measles, or severe pneumonia

"I favour COPD over bronchiectasis because Mrs. Fletcher has a classic 40 pack-year smoking history — the strongest risk factor for COPD — and the characteristic barrel chest with global hyperinflation rather than the segmental consolidation pattern of bronchiectasis. Her sputum is described as chronic and productive rather than the large-volume, frankly purulent sputum typical of bronchiectasis. Her symptoms follow a steadily progressive decline rather than the recurrent acute infective exacerbations more typical of bronchiectasis."

DIFFERENTIAL 2: LEFT VENTRICULAR FAILURE

SUPPORTING FEATURES

- Dyspnoea on exertion and at rest in a 74-year-old
- Peripheral oedema (ankle swelling)
- Fine bibasal crackles on auscultation
- Orthopnoea and paroxysmal nocturnal dyspnoea

"I favour COPD over left ventricular failure because the dominant picture is obstructive airflow limitation with emphysematous changes — barrel chest and air trapping — rather than the fluid overload and venous congestion of LVF. Her cough produces mucoid sputum, not frothy pink sputum. Her smoking history provides a much stronger aetiological explanation for her symptoms. However, given her age and smoking history, I would always screen for concurrent cardiac disease — COPD and ischaemic heart disease commonly coexist."

4 Objective Assessment & Special Tests

BASELINE OBSERVATIONS — ALWAYS RECORD FIRST

RR: expect 18–24/min at rest (tachypnoeic). SpO_2: aim ≥88–92% (hypoxic drive — do not target >95% without medical guidance). HR & BP. Accessory muscle use: SCM and scalenes visible at rest. Pursed-lip breathing — self-adopted PEEP to prevent dynamic

airway collapse. Breathing pattern: upper chest dominant, ↓ lateral costal expansion. Percussion: hyperresonance bilaterally confirms hyperinflation.

Test 1: 6-Minute Walk Test (6MWT)

WHAT IT MEASURES

Submaximal exercise capacity and functional endurance — reflects real-world activity tolerance. Gold standard for pulmonary rehab eligibility and outcome measurement.

HOW TO PERFORM

1. Flat, unobstructed 30m corridor; mark turnaround points clearly
2. Standardise instructions: "Walk as far as you can in 6 minutes at your own pace — you may slow down or rest, but do not stop the clock"
3. Monitor SpO_2 and HR continuously; Borg dyspnoea at baseline, 3 min, and 6 min
4. Record total distance in metres; note any desaturation — if SpO_2 drops below 85%, stop the test
5. Allow full rest before repeating if second test required (minimum 1 hour)

POSITIVE FINDING AND CLINICAL MEANING

<350m = significantly impaired functional capacity; correlates with poor prognosis and high rehab need. SpO_2 drop >4% from baseline = exercise-induced desaturation → ambulatory O_2 assessment required.

WHY I SELECT THIS TEST

"I select the 6MWT because it directly quantifies the functional deficit Mrs. Fletcher is experiencing in her daily life — reduced ability to walk and complete household tasks. It is the standard outcome measure for pulmonary rehabilitation and will determine whether supplemental O_2 is required during exercise."

Test 2: MRC Dyspnoea Scale

WHAT IT MEASURES

Grades breathlessness severity (1–5) based on the level of activity that provokes it. Validated, simple, and directly used to determine pulmonary rehab eligibility (Grade ≥2).

HOW TO PERFORM

1. Read each descriptor aloud to the patient
2. Grade 1: only breathless with strenuous exercise. Grade 2: hurrying on level ground or walking up a slight hill. Grade 3: walks slower than peers or stops for breath after 100m on level ground. Grade 4: too breathless to leave house or breathless when dressing. Grade 5: too breathless to leave the room
3. Patient selects the grade that best reflects their current experience
4. Record and use as a baseline outcome measure

POSITIVE FINDING AND CLINICAL MEANING

Expected Grade 3–4 — confirms significant dyspnoea-related disability and meets threshold for pulmonary rehabilitation referral.

WHY I SELECT THIS TEST

"I select the MRC scale because it directly answers the referral question — is Mrs. Fletcher suitable for pulmonary rehabilitation? It is quick, validated, and provides a grade I can track as an outcome measure at programme completion."

Test 3: Chest Auscultation and Wall Mobility Assessment

WHAT IT MEASURES

Air entry quality, breath sound character, adventitious sounds (wheeze, crackles); chest wall movement symmetry and breathing pattern.

HOW TO PERFORM

1. Patient seated, chest exposed with consent
2. Auscultate anterior, lateral, and posterior fields — minimum 6 zones, bilateral comparison
3. Assess lateral costal expansion: hands bilaterally on lower chest, thumbs meeting at midline — note symmetry and excursion on deep breath
4. Percussion: compare bilaterally — hyperresonance confirms hyperinflation
5. Observe breathing pattern: RR, I:E ratio, pursed-lip breathing, upper chest vs. diaphragmatic movement

AUSCULTATION FINDINGS EXPECTED

Globally reduced breath sounds (air trapping ↓ airflow velocity). **Expiratory wheeze** (dynamic airway collapse). Possible **coarse crackles** at bases if secretions present. **Prolonged I:E ratio** (>1:3) = obstructive pattern. ↓ Lateral costal expansion bilaterally.

WHY I SELECT THIS TEST

"I include chest auscultation because it directly informs airway clearance decisions — if coarse crackles are present, ACBT becomes a priority intervention. It also confirms the degree of obstruction and hyperinflation, guiding my breathing exercise prescription."

OUTCOME MEASURE: COPD ASSESSMENT TEST (CAT)

8-item questionnaire measuring COPD symptom burden and quality of life impact. Score 0–40; >10 = significant impact. Minimum clinically important difference (MCID) = 2 points. Use at baseline and post-rehabilitation to quantify change. Pairs with 6MWT distance for a comprehensive functional picture.

5 Treatment Plan

INITIAL PHASE — WEEKS 1–2

Airway Clearance: Active Cycle of Breathing Technique (ACBT)

How to do: Breathing control (relaxed tidal breathing, 3–4 breaths, shoulders down) → thoracic expansion exercises (3–4 slow deep breaths, optional 3-second inspiratory hold) → forced expiration technique / "huff" (open glottis, mid-to-low lung volume, not a cough) → repeat cycle 3–4 times → productive cough at end if needed.

REGRESSION: Breathing control + gentle assisted cough only if fatigued; sitting forward-lean position

→

PROGRESSION: Add gravity-assisted drainage positions; introduce Flutter/Acapella oscillating PEP device; increase cycles to 4–5

Breathing Exercise: Pursed-Lip Breathing (PLB) + Diaphragmatic Re-education

PLB how to: Inhale through nose 2 counts; exhale slowly through pursed lips ("blow out a candle slowly") 4 counts. Creates back-pressure (intrinsic PEEP) → slows expiratory flow → prevents dynamic airway collapse → ↓ RR and ↑ tidal volume.
Diaphragmatic how to: Semi-reclined or supine; hand on abdomen; breathe to push hand out on inspiration, not chest. Aims to recruit diaphragm rather than accessory muscles.

REGRESSION: PLB only in sitting; 2:3 ratio; shorter sessions →

PROGRESSION: PLB during walking and stair climbing; extend to 2:6 ratio; practise during 6MWT warm-up

Exercise / Mobility: Low-Intensity Walking Programme

Starting level: 10–15 minutes self-paced walking, 3x/week. Target Borg dyspnoea 3–4/10. Monitor SpO_2 — maintain ≥88%. Rest-to-walk ratio guided by symptoms.

REGRESSION: Seated lower limb exercises (leg raises, ankle pumps, sit-to-stand ×5) →

PROGRESSION: 15 → 20 → 30 minutes; introduce incline; add resistance band upper limb work (unsupported arm exercise trains ventilatory muscles)

PATIENT EDUCATION — KEY MESSAGES

- **Energy conservation and pacing:** Plan activities across the day — prioritise and delegate. Stop at 70% capacity rather than waiting until exhausted and requiring long recovery.
- **Smoking cessation:** If still smoking — the single most effective intervention; refer to cessation service. Lung function decline slows markedly after stopping.

- **Recognising and acting on exacerbations early:** Increase in sputum colour/volume, worsening breathlessness → contact GP early. A written COPD action plan is essential.
- **Correct inhaler technique:** Review with spacer if using pMDI — poor technique means poor drug delivery, meaning symptoms remain unnecessarily severe.

PROGRESSION CRITERIA — MEASURABLE TARGETS BEFORE ADVANCING

SpO_2 ≥88% throughout exercise session | Borg dyspnoea ≤4/10 at session end | RR <25/min at peak exercise | Walking duration ≥20 continuous minutes | CAT score improvement ≥2 points from baseline

LONG-TERM — WEEKS 4–12: PULMONARY REHAB AND SELF-MANAGEMENT

- **Formal pulmonary rehabilitation programme:** Twice-weekly supervised sessions (exercise + education) for 6–8 weeks. Strong evidence for ↑ 6MWT distance, ↓ dyspnoea, ↓ hospital admissions, and ↑ quality of life.
- **Resistance training:** Lower limb focus (quads, glutes) — provides greatest functional gains in COPD. Unsupported upper limb exercise (arms above shoulder height) trains ventilatory muscles — progress gradually.
- **Self-management:** Written COPD action plan; home walking diary; buddy walking scheme or community group for social support and accountability.
- **Discharge criteria:** SpO_2 stable during ADLs; 6MWT improvement ≥30m from baseline (MCID); able to self-manage acute exacerbations per action plan; independent home exercise programme established.

Examiner Demonstration — Techniques to Show in the Room

1. **ACBT cycle:** Demonstrate breathing control (relaxed, shoulders down, tidal breath) → 3 thoracic expansion breaths (feel the chest open) → huff at mid-lung volume (open glottis, not a cough — "like steaming up a mirror"). Narrate each step and explain the physiology as you go.
2. **Pursed-Lip Breathing:** Demonstrate the 2:4 ratio. Coach the examiner as the patient — "Breathe in slowly through your nose... now blow out gently through your lips as if you're cooling hot soup. Don't force it." Explain that it creates backpressure to prevent airway collapse.

Ms. Elena Rodriguez

CASE B

24F, Postgraduate student | Acute asthma exacerbation | Triggered by viral URTI + high pollen

Breathless at rest | Speaking in short phrases | Accessory muscle use | Audible expiratory wheeze | Blue reliever not providing relief

Setting: ED post-stabilisation — assess breathing pattern and provide inhaler technique education

1 Subjective Assessment

"Hello Ms. Rodriguez, I'm [your name], a physiotherapist. I can see you're still finding breathing difficult and I want to help. I'll ask a few short questions — you don't need to speak in full sentences, just nod or use a few words. I'll keep checking in on how you're feeling. Is that okay?"

⚠ CONFIRM MEDICAL STABILITY BEFORE PROCEEDING

Verify SpO_2 ≥92% on current O_2 therapy. Confirm initial bronchodilator (salbutamol nebuliser) has been administered. Confirm no deterioration since ED arrival. Physiotherapy is adjunctive to — not a replacement for — medical management. If in doubt, check with the medical team first.

KEY QUESTIONS TO ASK (6) — USE CLOSED/SHORT-ANSWER QUESTIONS

- **1. Is your breathing better, same, or worse than when you arrived?** *— I ask this first because it tells me whether she is improving or deteriorating. If worsening, I escalate medical care before any physio assessment continues.*
- **2. When did this episode start, and what triggered it?** Cold, pollen, exercise, stress? *— Viral + atopic triggers suggest eosinophilic, IgE-mediated asthma. Identifying triggers is essential for long-term self-management and avoidance strategies.*
- **3. How many puffs of your blue inhaler did you use today, and when was the last dose?** *— More than 4 puffs of SABA in 24 hours signals poor control. Ineffective relief points either to poor technique (most likely) or to severity overwhelming bronchodilation — both are actionable.*
- **4. Do you have a preventer inhaler — and do you take it every day?** *— ICS preventer adherence is the number one modifiable factor in asthma control. Non-adherence is the most common reason for ED presentation and is directly addressed by physiotherapy education.*
- **5. Have you ever been admitted to ICU or put on a breathing machine for your asthma?** *— Previous near-fatal asthma is the strongest predictor of future near-fatal episode. This changes my urgency, the depth of education I provide, and the follow-up plan.*
- **6. Any tightness in the throat rather than the chest? Any heart racing?** *— Throat tightness suggests vocal cord dysfunction (VCD), which can coexist with asthma and does*

not respond to bronchodilators — important differential. Palpitations may be salbutamol side-effect or anxiety.

RED FLAGS (5)

⚠ Silent Chest

No wheeze despite severe SOB — air movement so reduced it cannot generate wheeze. Pre-respiratory arrest

→ EMERGENCY — call resuscitation team immediately; do not proceed

⚠ SpO_2 <92% / cyanosis

Severe hypoxaemia from V/Q mismatch; impending respiratory failure

→ Escalate O_2; urgent medical review; prepare for possible intubation

⚠ Confusion / altered GCS

CO_2 narcosis from hypercapnic respiratory failure — life-threatening

→ Emergency intubation likely required; call team immediately; stop physio

⚠ RR >30 breaths/min

Severe respiratory distress; respiratory muscle fatigue imminent; rapid deterioration possible

→ Do not proceed with physio; escalate medical management now

⚠ Pulsus paradoxus >10 mmHg

Severe airflow obstruction causing large intrathoracic pressure swings with each breath

→ Indicator of severe acute asthma; urgent medical escalation required

PROBLEM LIST

PHYSICAL IMPAIRMENTS	FUNCTIONAL ACTIVITY LIMITATIONS
Acute bronchoconstriction — ↑ airway resistance, ↓ expiratory flow	Unable to speak in full sentences
Dysfunctional breathing pattern — upper chest, accessory muscle dominant	Cannot perform ADLs; completely activity-limited
Possible mucus hypersecretion and plugging	Unable to attend academic commitments
Suboptimal inhaler technique — reliever not working	Anxiety and breathlessness-fear cycle perpetuating symptoms
Likely poor ICS preventer adherence — poorly controlled baseline	Risk of recurrent ED presentations without improved self-management

2 Pathophysiology

WHAT IS HAPPENING IN THE AIRWAYS

- Asthma is characterised by **chronic eosinophilic airway inflammation** and bronchial hyperresponsiveness (BHR). In sensitised individuals (atopic asthma), airway epithelium is primed to over-react to triggers. Ms. Rodriguez has two simultaneous triggers: viral URTI causes direct epithelial injury, releasing inflammatory mediators (histamine, leukotrienes, prostaglandins) that amplify BHR; pollen allergens cross-link IgE on mast cells → degranulation → acute bronchoconstriction.
- Bronchoconstriction dramatically increases airway resistance. By Poiseuille's law, resistance $\propto 1/r^4$ — halving the airway radius increases resistance 16-fold. Expiration is disproportionately affected because airways lose their radial traction support during expiration, narrowing further. This generates the characteristic **audible expiratory wheeze**.
- **Air trapping** occurs as gas distal to obstructed airways cannot fully escape. Acute dynamic hyperinflation results — the diaphragm flattens, forcing accessory muscle recruitment and dramatically increasing the work of breathing. This is why Ms. Rodriguez is visibly using her neck muscles.
- Mucous gland hypertrophy and goblet cell hyperplasia from chronic inflammation produce excess, viscous secretions. These may form **mucus plugs**, causing regional airway occlusion, worsening obstruction, and creating V/Q mismatch → hypoxaemia despite normal RR.
- Hyperventilation (driven by hypoxia and anxiety) reduces $PaCO_2$. Paradoxically, airway hypocapnia causes bronchoconstriction — further worsening the obstruction. This creates a vicious cycle that breathing retraining can directly interrupt.
- Salbutamol (SABA) acts on β_2 receptors → smooth muscle relaxation → bronchodilation. Ineffective relief in Ms. Rodriguez most likely reflects **poor inhaler technique** — the most common cause — rather than medication failure per se.

WHY MS. RODRIGUEZ FITS THIS PATHOPHYSIOLOGY

24F with likely atopic background (pollen trigger = IgE-mediated sensitisation). Viral URTI provides the second-hit amplification of airway inflammation. Audible expiratory wheeze confirms bronchoconstriction with dynamic airway collapse during expiration. Short phrases = ↑ work of breathing consuming ventilatory reserve. Ineffective SABA = almost certainly poor pMDI technique (requires precise press-and-breathe coordination). This is acute-on-chronic, poorly controlled asthma.

3 Differential Diagnoses

DIFFERENTIAL 1: VOCAL CORD DYSFUNCTION (VCD) / INDUCIBLE LARYNGEAL OBSTRUCTION (ILO)

SUPPORTING FEATURES

- Young female — VCD is most prevalent in young women aged 20–40
- Episodic breathlessness — can be triggered by stress or exercise (postgraduate student, high pressure environment)
- Does not fully respond to bronchodilators
- Stridor or predominantly inspiratory wheeze rather than expiratory
- Symptoms may resolve with speech therapy or during sleep

"I favour acute asthma over VCD because Ms. Rodriguez has a known asthma diagnosis, dual atopic triggers (viral + pollen — not typical VCD triggers), and the wheeze is audibly expiratory in character rather than inspiratory. However, I am aware that VCD coexists with asthma in up to 40% of patients, and I would screen for it as part of long-term management — particularly given her stress exposure as a postgraduate student."

DIFFERENTIAL 2: ANAPHYLAXIS WITH BRONCHOSPASM

SUPPORTING FEATURES

- Atopic individual with acute onset severe breathlessness
- High-pollen allergen exposure on day of presentation
- Bronchospasm can be a component of a systemic allergic reaction
- In severe anaphylaxis: urticaria, angioedema, hypotension would also be present

"I favour acute asthma exacerbation over anaphylaxis because the presentation is isolated to the respiratory system without systemic allergic features — no urticaria, no angioedema, no hypotension. The gradual onset in the context of a viral illness plus pollen exposure is more consistent with asthma than the typically rapid, multisystem reaction of anaphylaxis. The medical team has not administered adrenaline, which would have been the immediate priority if anaphylaxis were suspected."

4 Objective Assessment & Special Tests

BASELINE OBSERVATIONS — ALWAYS DOCUMENT

RR: expect 20–30/min (tachypnoeic). SpO_2: must be ≥92% before proceeding. HR: often elevated (salbutamol effect + sympathetic drive). Observe: SCM and scalene use, intercostal recession, tracheal tug, nasal flaring. Speaking pattern: short phrases only = moderate-severe. Anxiety level: will impact breathing pattern assessment — acknowledge and address early.

Test 1: Breathing Pattern Assessment

WHAT IT MEASURES

Identifies dysfunctional breathing patterns — upper chest dominance, hyperventilation, irregular rhythm, mouth breathing — that worsen bronchoconstriction and maintain the acute episode. Directly guides the breathing retraining intervention.

HOW TO PERFORM

1. Observe patient at rest — note chest vs. abdominal movement (place one hand on chest, one on abdomen)
2. Count RR and observe rhythm — note frequent sighs, irregular breathing, breath-holding
3. Note mouth vs. nasal breathing (mouth breathing → cold, dry air → ↑ bronchoconstriction)
4. Observe I:E ratio — in asthma, expiration should be prolonged; if inspiration-dominant, consider hyperventilation or VCD
5. Note paradoxical abdominal movement (inward on inspiration) = diaphragmatic dysfunction

POSITIVE FINDING AND CLINICAL MEANING

Upper chest dominant breathing, elevated RR, mouth breathing, frequent sighs = dysfunctional breathing pattern. This causes hyperventilation → ↓ $PaCO_2$ → airway hypocapnia → direct bronchoconstriction — worsening the acute episode.

WHY I SELECT THIS TEST

"I select breathing pattern assessment because post-stabilisation, dysfunctional breathing is both a consequence and a perpetuator of the acute episode. Identifying and correcting it with controlled breathing techniques is the primary physiotherapy contribution in this acute scenario — and this assessment directly informs that prescription."

Test 2: Inhaler Technique Assessment (Observed)

WHAT IT MEASURES

Identifies specific, correctable errors in pMDI or DPI technique — a modifiable cause of poor asthma control. Up to 70–80% of patients demonstrate incorrect technique.

HOW TO PERFORM

1. Ask patient to demonstrate using their own device (or a placebo pMDI)
2. Observe against the standardised checklist: shake canister; remove cap; exhale gently; seal lips; begin slow inhalation; actuate canister simultaneously; continue slow breath over 3–5 seconds; hold breath 10 seconds; wait 30–60 seconds before second puff
3. Note specific errors: actuation before breath begins; actuation after breath is complete; excessively fast inhalation rate; inadequate breath hold; not exhaling before use; lips not sealed around mouthpiece

4. For DPI (Turbohaler, Accuhaler): technique is different — sharp, fast inhalation required, no shaking, do not exhale into device
5. Document errors observed; correct one at a time with demonstration and return demonstration

POSITIVE FINDING AND CLINICAL MEANING

Poor coordination (actuation before breath or too late), fast inhalation rate, inadequate breath hold — these mean <10% of drug reaches the lungs. Correcting technique achieves the same therapeutic benefit as stepping up medication.

WHY I SELECT THIS TEST

"I select inhaler technique assessment because it is the most likely explanation for why Ms. Rodriguez's blue reliever is not providing relief. It is directly actionable — I can identify and correct specific errors in this appointment. It is also the most impactful educational intervention I can offer before discharge."

Test 3: Chest Auscultation (Post-Bronchodilator)

WHAT IT MEASURES

Assesses response to bronchodilator therapy; monitors for silent chest (emergency indicator); identifies any secretion presence requiring airway clearance.

HOW TO PERFORM

1. Patient seated upright if tolerated; expose chest with consent
2. Auscultate all zones bilaterally; compare left to right
3. Note character: wheeze (inspiratory vs. expiratory), crackles, reduced air entry
4. Assess for improvement compared to initial presentation
5. Any zone of silence = emergency — escalate immediately

AUSCULTATION FINDINGS EXPECTED

Bilateral expiratory wheeze — polyphonic (multiple airway sizes involved). **Reduced air entry** in severe obstruction. Post-treatment: ↓ wheeze, ↑ air entry = responding to bronchodilators. **Coarse crackles** if mucus plugging present → consider ACBT. Silent chest in any zone = immediate escalation.

WHY I SELECT THIS TEST

"I select auscultation to confirm current obstruction level, monitor bronchodilator response, and detect any secretion component requiring airway clearance intervention. Critically, it allows me to identify a silent chest early — the most dangerous sign in acute asthma."

OUTCOME MEASURE: ASTHMA CONTROL TEST (ACT)

5-item validated questionnaire. Score ≤19 = uncontrolled asthma; ≤15 = very poorly controlled. MCID = 3 points. Use at discharge and at GP follow-up. Pairs with peak flow

diary (target ≥80% predicted or personal best). Quick, self-administered, and highly relevant to Ms. Rodriguez's education needs.

5 Treatment Plan

PRIORITY ORDER IN THE ACUTE SETTING

1. Confirm medical stabilisation and SpO_2 ≥92% → 2. Optimal positioning → 3. Controlled breathing retraining → 4. Inhaler technique education → 5. Self-management and action plan. Do NOT commence any exercise until SpO_2 stable ≥92% and RR <25/min.

INITIAL PHASE — ACUTE MANAGEMENT (ED / HOURS 1–48)

Breathing Exercise: Controlled Breathing for Acute Asthma

Position first: High side-lying or forward-lean sitting (arms resting on pillow/table) — reduces accessory muscle load and allows the diaphragm to descend effectively.
How to do: Slow nasal breathing in (nasal breathing warms, humidifies, and filters air — reduces bronchial irritation). Gentle, prolonged expiration through nose or slightly open mouth — do not force. Target I:E ratio 2:4. Cue: "Breathe in through your nose slowly... now breathe out gently and slowly — don't push it out, just let it go."

REGRESSION: Positioning + reassurance only if too distressed; coach one slow breath out at a time; guided imagery ("imagine blowing out a single candle slowly")

→

PROGRESSION: Introduce lateral costal expansion breathing; increase session duration; practise during short walks before discharge

Inhaler Technique Education: pMDI with Spacer

Correct pMDI technique (7 steps):

1. Shake the inhaler; remove the cap
2. Breathe out gently (not forcefully)
3. Seal lips firmly around mouthpiece (or spacer mouthpiece)
4. Begin slow inhalation — press canister down once at the very start
5. Continue slow inhalation over 3–5 seconds
6. Hold breath for 10 seconds (maximises drug deposition)
7. Wait 30–60 seconds before taking the next puff

Spacer device: Strongly recommend. Removes need for perfect coordination; increases lung deposition from ~10% to ~20–40%. Demonstrate attaching, shaking through spacer, and using one puff per actuation with spacer.

REGRESSION: Demonstrate on placebo inhaler only while patient observes; use verbal description and diagram

→

PROGRESSION: Patient demonstrates return demonstration independently; review DPI technique if relevant; practise at follow-up

Exercise / Mobility (Post-Stabilisation Only)

Starting level: Bedside mobility only — sitting over edge of bed, standing transfers with rest. No aerobic exercise until SpO_2 ≥92% stable and RR <24/min.

REGRESSION: Breathing exercises in bed only; positioning for ventilation →

PROGRESSION: Short corridor walks with controlled breathing; discharge to community follow-up with self-management plan

PATIENT EDUCATION — KEY MESSAGES

- **Preventer vs. reliever — the most important distinction:** "Your preventer inhaler [ICS] must be taken every day — even when you feel completely well. It reduces the inflammation that causes these attacks. Your blue inhaler is for relief in the moment — it does not treat the underlying inflammation."
- **Written Asthma Action Plan:** Traffic-light system — Green (well, take preventer as usual); Amber (increasing symptoms — take extra reliever, start oral steroid if prescribed, contact GP); Red (severe attack not responding — call 999 immediately). Every patient should leave hospital with a written plan.
- **Trigger avoidance:** Check daily pollen count (high pollen days — keep windows closed, shower after being outdoors, wear wrap-around sunglasses). For viral illness — begin reliever use early at first sign of symptoms; contact GP proactively.
- **When to call emergency services:** "If your blue inhaler is not working after 4 puffs in 10 minutes, your breathing is getting worse, or you cannot speak — call 999. Never wait to see if it improves on its own in that situation."

PROGRESSION CRITERIA — BEFORE DISCHARGE OR STEP-UP

SpO_2 ≥92% on room air | RR <20/min at rest | Speaking in full sentences | Wheeze resolved or minimal on auscultation | Borg dyspnoea ≤3/10 at rest | Peak flow ≥75% predicted or personal best

LONG-TERM — WEEKS 1–4 POST-DISCHARGE AND SELF-MANAGEMENT

- **GP/Asthma nurse follow-up within 48 hours of discharge:** Review medication step; ensure written action plan provided; consider specialist referral if this is a second ED attendance within 12 months.

- **Breathing retraining programme:** Buteyko or Papworth method — evidence supports ↓ reliever use and ↑ quality of life in chronic dysfunctional breathing. Particularly appropriate given the likely anxiety-asthma overlap.
- **Psychological support:** Screen for health anxiety and panic disorder (both share positive feedback loops with breathlessness). Refer for CBT if indicated — stress management also reduces exacerbation frequency.
- **Discharge criteria:** Demonstrates correct inhaler technique independently; has personalised written action plan; peak flow ≥80% personal best; ACT score ≥20 (well-controlled); follow-up appointment within 1 week confirmed.

Examiner Demonstration — Techniques to Show in the Room

1. **Forward-lean position + controlled breathing:** Sit forward, arms resting on thighs or on a table in front. Demonstrate slow nasal inhalation with a gentle, prolonged expiration. Coach the examiner as the patient — use a calm, measured pace. Explain: "This position reduces the work of the accessory muscles and allows the diaphragm to drop more effectively."
2. **pMDI technique with spacer:** Walk through all 7 steps on a placebo inhaler. Attach spacer and explain: "The spacer creates a holding chamber so Ms. Rodriguez doesn't need to perfectly coordinate pressing and breathing — and it dramatically increases the amount of medication that actually reaches the lungs."